

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Mid Term Examination, 2024-25 (NEP)

PHYSICS (SEC)**Paper Code: PHYSE21 : Energy Conversion and Storage for Sustainable Development**

Time: 1 Hour

Maximum Marks: 20

Instructions: Answer all questions.

1. Describe Biochemical energy conversion, and Solar energy conversion technologies. [4]
2. Briefly describe the two major categories of thermal energy storage technologies. [3]
3. In a water storage tank containing 1 metric ton of water, temperature of the water initially was 28 °C and final temperature is 50 °C. How much heat energy is stored? [3]
4. Assume that you have five 100W lamps, ten 40W tube-lights, five 100W ceiling fans, one 400W refrigerator, and one 200W TV in your house.
 - (a) What is the minimum electricity load requirement for your house?
 - (b) Assume further that your electricity provider charges 5 per electricity unit. If all your appliances run at 50% capacity about 8 hours a day on average, how much should be your monthly electricity bill?
 - (c) What is your annual electricity consumption in electricity units?
 - (d) Suppose that you can buy solar panel from the market in multiples of 1 kW panels and it costs 70,000 for each 1 kW panel. Assuming a 1 kW panel would produce 1400 electricity units on average in a year, how much money will you save per year if you install 2kW solar panels and in how many years you are going to recover your upfront installation cost?
 - (e) The PM Surya Ghar Yojana currently provides subsidy upto 78,000 (30,000 per kW up to 2 kW and 18,000 per kW for additional capacity up to 3 kW). Consider the case where you receive the eligible subsidy and calculate in how many years you will break even in this case.
 - (f) Would your savings increase if your house was in Jaipur instead of Varanasi?[6]
5. Under National Green Hydrogen Mission, how does India aim to source green Hydrogen? Name two domestic demands where green Hydrogen are expected to be utilized? [2]
6. (a) What are the broad objectives of the Standards & Labelling Scheme of BEE?

- (b) State True or False: *Voltas's model-XYZ Air-conditioner received 4 star rating from BEE in 2024. In 2025, the same model received 5 star.*

[3]